A Flip Is Not a Faithfulness Signal: Four Controls on FAITHLM and a Vietnamese Stress Test

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Abstract

FAITHLM scores an explanation by negating it and measuring how much the target model's prediction shifts. We reproduce its COPA setting with the paper's own models, obtaining 0.872 against the ~ 0.85 reported, and then subject the metric to four controls it does not include. Three attack *validity*: an off-topic hint already flips the target on 55.4% of instances; a hint written for a different question flips it on 14.0%; and a target whose answers are fully explained by "pick the first-listed option"—$83/117$ correct and $83/117$ first-picks are the same 83 instances—still scores 0.755. One attacks *reliability*: redrawing the contrary hint a single time, with the explanation, question and temperature unchanged, loses more than a third of all flips (0.647, $[0.596, 0.694]$). Across six explainer $\times$ target cells the score never leaves 0.930–0.975 and no comparison is significant, so the noise inside one measurement is an order of magnitude larger than the gaps the metric is asked to resolve. We trace much of this to the released implementation reading predictions by regular expression over generated text, which discards 45–47% of Vietnamese predictions; reading choice probabilities instead—as the paper's own formulation specifies—raises the flip rate from 0.374 to 0.509 ($p = 0.00011$).

1 Introduction

Explanations produced by large language models are fluent but need not describe the computation that produced the answer (Jacovi and Goldberg, 2020; Turpin et al., 2023). FAITHLM (Chuang et al., 2026) proposes to settle this by intervention: an explainer LLM writes a *contrary hint* $\neg E_{\text{NL}}$ asserting the opposite of the explanation, and fidelity is the resulting shift,

$$S_E := f(X) - f(X \mid \neg E_{\text{NL}}). \tag{1}$$

A large shift is read as evidence that the explanation carried decision-relevant content.

We reproduce this measurement and then ask what a large shift actually licenses. Our contributions:

1. **Reproduction** of the COPA setting with the paper's own target and explainer (§6.1).
2. **Three validity controls** showing the score survives removing the explanation's content, and is high even for a target that does no reasoning (§6.3).
3. **A reliability control** showing a third of all flips are properties of a single sampling draw rather than of the explanation (§6.4).
4. **A concrete cause and fix**: the released code reads predictions from prose, not from the model's distribution (§6.5).
5. **A per-language XCOPA evaluation**. The original transfers COPA to XCOPA without naming a language; we report Vietnamese explicitly (§6.6).
6. **An audit of the trigger-prompt optimiser**, which in one of three runs never beat its human-written seed and in another scored the client's error string as a candidate prompt (§6.7).

2 Related Work

Faithfulness is commonly probed either by *simulatability*—can a model reproduce predictions from an explanation (Madsen et al., 2024)—or by perturbing the input. Jacovi and Goldberg (2020) argue these test different properties. Chain-of-thought text is demonstrably unfaithful under controlled manipulation (Turpin et al., 2023; Lanham et al., 2023). FAITHLM avoids token deletion, which is ill-defined when the decision-relevant content is latent, by intervening on meaning. Our concern is not that move but the absence of a null condition, and the reliability of a score read from a single stochastic draw.

3 Dataset

Balanced COPA. Roemmele et al. (2011) pose binary causal commonsense questions; Kavumba et al. (2019) add mirrored instances to suppress lexical shortcuts. We use the 500-item test split.

XCOPA-vi. Ponti et al. (2020) translate COPA into 11 languages with professional translators. The Vietnamese test split contains *the same* 500 *items* as the English split, verified item by item, so the pair is parallel and language is the only variable. Neither corpus carries gold explanations, which is why we cannot report the original's second metric (§8).

4 Method

The measurement. The target answers the question; an explainer proposes E_{NL} from the (question, answer) pair alone—it never sees the target's own generation, which the pipeline discards after parsing, and its instruction literally asks it to *guess* the reason; the explainer then writes $\neg E_{\mathrm{NL}}$; the target answers again with the hint prepended. An optimisation loop rewrites the explanation to maximise Eq. 1 and stops at the first non-zero score.

Validity controls. **(A) Off-topic hint**: a hint unrelated to the question ("The weather forecast mentions scattered clouds tomorrow afternoon."). **(B) Mismatched hint**: the contrary hint written for a *different* instance—same register, length and form, differing only in which question it belongs to, so it cannot be dismissed as trivially ignorable. **(C) Position bias**: we test whether a target's accuracy is explained by always choosing the first-listed option, and if so, what fidelity that target still earns.

Reliability control (D). Keeping the winning explanation fixed, we redraw the contrary hint once at the same temperature and re-ask the target, scoring with the pipeline's own rule. If the score is a property of the explanation, it should reproduce.

Score modes. The released code computes $|\Delta\text{accuracy}|$ over an argmax answer parsed out of generated text. We add PROB_ACCURACY (argmax over choice probabilities) and LOGPROB (the continuous probability shift), both keeping Eq. 1 and changing only how f is read.

5 Experimental Setup

For the reproduction we follow Table 2 of the original: 20 steps, target temperature 0.7, explainer temperature 0.9, top-p 0.9, 500 instances, target `phi-2` and explainer `gpt-3.5-turbo` (Javaheripi and Bubeck, 2023). Both of the original's explainers have since been retired, so a separate axis substitutes four current ones with everything else fixed, using the paper's English prompts verbatim (diffed against the released commit; only whitespace differs). Additional targets—Qwen3.5-4B and DeepSeek-v4-flash—probe the Vietnamese setting. All runs, per-instance outputs and code are released.

Three implementation details materially affect any reproduction. **(i)** `do_sample` is never set, so `transformers` ignores the temperature the paper specifies and decodes greedily; we verified both regimes give the same result, but a run following Table 2 is not running the released configuration. **(ii)** The early-stopping condition is tested only every fifth iteration, so 16.6% of executed iterations occur after the criterion is already met. **(iii)** 2026-generation models are reasoning models: one target spent 179 of a 200-token budget on reasoning and returned empty content, and one explainer spent 827 reasoning tokens on a five-token answer. Disabling reasoning eliminated empty completions across 5,924 calls and also brings these models closer to the paper's non-reasoning originals.

6 Results

6.1 The reproduction succeeds

With Phi-2 and GPT-3.5 on Balanced COPA, fidelity reaches 0.872 over 500 instances against the ~ 0.85 the original reports. A modern explainer under the same protocol gives 0.860. The measurement reproduces.

6.2 The metric is saturated

The contrary hint flipped *every* parseable prediction in the reproduction: all 354 correct and all 34 incorrect answers score exactly 1.0. Across a full 3×2 explainer $\times$ target grid ($N = 200$ per cell) the score never leaves 0.930–0.975; pooled by target the two are indistinguishable (0.943 vs 0.950, $z = -0.51$, $p = 0.607$), and no explainer keeps its rank across targets. Among four explainers from four laboratories, all six pairwise z-tests are non-significant ($p = 0.216$–0.754). Doubling N from 100 to 200 tightened the standard error from $\sim$2.0% to 1.5–1.8% and still found nothing.

	COPA-en	XCOPA-vi
Fidelity	0.860	0.336
Control A (off-topic)	0.554	0.122
Corrected	**+0.306**	**+0.214**

Table 1: Control A, 500 instances each, Phi-2 target. Roughly 64% of the English score survives when the hint has nothing to do with the question.

Arm	Flips	Repro.	95% CI
Gemini-3.5-flash	178	0.607	[0.533, 0.676]
GPT-5.6-luna	187	0.684	[0.615, 0.747]
Pooled	**365**	**0.647**	[0.596, 0.694]

Table 2: Control D. One redraw of the contrary hint, everything else fixed.

6.3 Validity: the score is not specific to the explanation

Table 1 reports Control A. Control B, designed independently on a different target and explainer, agrees: an own-question hint flips 0.370 of instances against 0.140 for a hint belonging to another question ($p < 0.0001$, $n = 200$). Two controls differing in design, target and explainer land in the same place—baselines of 0.122 and 0.140, corrected effects of +0.214 and +0.230. The effect is real and semantic; roughly one flip in seven is noise, and on English about two thirds of the headline number is suggestibility.

Control C is sharper. On XCOPA-vi with English prompts, Phi-2 parses 117 of 200 instances; among those, the 83 it answers correctly are *the same* 83 on which it picks the first-listed option. A bot that always picks first scores 62% on this subset, leaving a 9-point comprehension margin. This target is not reasoning—and it is still scored at 0.755 fidelity, with explanations that never mention position. By construction those explanations cannot be faithful, yet the metric rates them highly.

6.4 Reliability: a third of flips do not reproduce

Table 2 is the result we consider most consequential. Holding the explanation, question, target and temperature fixed and redrawing the contrary hint *once*, only 64.7% of flips reproduce—far below a 0.90 stability threshold ($z = -16.1$) and significantly below 0.70 ($p = 0.026$). The two arms do not differ ($p = 0.12$), so this is a property of the measurement, not of an explainer. A reported 0.930 therefore means "flipped on one draw", not

Score mode	Flip rate	95% CI
ACCURACY (released)	0.374	[0.315, 0.432]
PROB_ACCURACY	0.460	[0.400, 0.520]
LOGPROB	**0.509**	[0.449, 0.570]

Table 3: XCOPA-vi, 500 instances. McNemar against the released mode: $p = 0.00011$ (LOGPROB), $p = 0.01003$ (PROB_ACCURACY).

"93% of explanations are faithful"; the two-draw-consistent rate is ≈ 0.564.

This also answers the obvious objection to §6.2—run more questions. The noise lives inside each single-instance measurement, ~ 35 points of it, against inter-explainer gaps of ≤ 3.5 points. Increasing N narrows the error bar of a quantity that is itself unstable.

Flips found on the first iteration reproduce better than those found later (0.715 vs 0.609 pooled, $p = 0.041$), which is the signature one would expect if the optimisation loop is partly harvesting lucky draws. We report this as suggestive only: it is significant on one arm and not the other, and would not survive correction for multiple comparisons.

6.5 A cause, and a fix

The released prompt ends at `### Response:` without stating an output format, then searches the continuation for `[choice]. . . @`. On XCOPA-vi this discards 45–47% of predictions, at the same rate for two very different targets—a prompt–parser mismatch, not a target failure. Among surviving predictions the parser disagrees with a normalised argmax on 59% of 150 audited instances. The metric also measures Δcorrectness rather than Δprediction: on one instance the hint flipped the answer outright and the released metric recorded 0.0, because neither answer matched the gold string.

Table 3 shows what changes when f is read from probabilities, as §3.1 of the original specifies. The gain is significant, but the informative comparison is PROB_ACCURACY against LOGPROB: $p = 0.103$ at $n = 500$, and since 500 is the entire test split this is a ceiling, not a sample-size limit. *Most of the improvement comes from abandoning the text parser, not from the continuous signal.*

6.6 On Vietnamese, the target dominates

Table 4 shows an 80-point accuracy range across targets, far exceeding any effect the metric reports.

Target	Prompt	Acc.	Unp.	Fid.
Phi-2	VI	11.8%	45.4%	0.336
Phi-2	EN	41.5%	41.5%	0.755
Qwen3.5-4B	EN	90.5%	6.5%	0.945
DeepSeek-v4-flash	EN	92.5%	2.0%	0.930

Table 4: XCOPA-vi by target and prompt-scaffold language.

Run	Seed	Best	`"API error."`
DeepSeek-v4-pro	0.467	0.533	5/16
DeepSeek-v4-flash	0.467	**0.400**	1/16
+ target reasoning	0.333	0.600	0/16

Table 5: Algorithm 2 on XCOPA-vi: 15 optimisation steps, 15 sampled instances per step. *Seed* is the human-crafted trigger prompt; *Best* is the highest score any optimised prompt reached afterwards.

Fidelity tracks the *parse rate* ($41.5\% \rightarrow 6.5\% \rightarrow 2.0\%$ unparsed against $0.755 \rightarrow 0.945 \rightarrow 0.930$) more closely than it tracks model capability, and the curve is not monotone in model strength. Phi-2's Vietnamese accuracy is not evidence of anti-correlated reasoning: Control C shows it is position bias, and under probability reading the same model sits at 0.520 against a majority-class floor of 0.530—chance, with an 87% preference for the first option.

A further variable is uncontrolled by the original's specification. With English instructions over Vietnamese content, the paper's `exp_instruction` never names an output language, and explainers disagree: measured by diacritic detection, explanation language ranges from 0.4% to 75.3% Vietnamese across explainers while fidelity stays inside 0.930–0.975. The metric cannot see a difference that coarse.

6.7 The trigger-prompt optimiser does not improve on its own seed

The original's second algorithm optimises the *trigger prompt* that elicits explanations, reporting that optimised prompts outperform the human-crafted seed. We ran it three times on XCOPA-vi (Table 5). In one run fifteen optimisation steps never beat the seed. The trajectories oscillate without trend and every run ends below its own best, so the returned prompt is not the best one found.

Two properties of the setup explain why this is expected rather than surprising. Each step scores on 15 sampled instances, so a score can only take values $k/15$ and the entire observed range spans a handful of instances. And because the sample is redrawn each step, consecutive scores are not comparable—the optimiser is being shown a noisy signal and asked to hill-climb on it, which is the same reliability problem as §6.4 at the level of the prompt rather than the instance.

One run also exposes an error-handling defect with direct consequences for the result. The explainer client returns the literal string `"API error."` on failure; the loop stores it as the next trigger prompt and scores it. Five of sixteen records in that run carry `"API error."` as their prompt, *including the final returned optimum*. A failed call is not distinguishable from a candidate prompt anywhere in the recorded trajectory.

6.8 Grounding the explainer in the target's reasoning

Because the explainer sees only the question and the parsed answer, its explanation is a hypothesis about *any* plausible reason for that answer—two models emitting the same choice receive the same explanation. Forwarding the target's own generated reasoning instead drops fidelity from 0.905 to 0.425 ($n = 200$) while iterations rise from 4.87 to 10.20. The direction is not self-interpreting: an explanation that states a strong, specific, wrong claim is easy to invert into a derailing hint, whereas one tracking the target's actual reasoning offers less leverage. Under a metric rewarding large shifts, being more faithful can score lower. Separating the readings needs the truthfulness metric, which our corpora cannot support.

7 Error Analysis

Of 14 errors across both English runs, 13 are parsing failures and one is a genuinely wrong choice; what the pipeline calls accuracy is closer to a parse-success rate. The failure cascades: when the answer is unresolvable the pipeline passes the literal string `X` to the explainer, which explains it ("*The model chose answer X based on the content of the question...*"), and the hint and score that follow are about an answer never produced. This is why unparseable instances show a corrected effect of only +0.018–+0.044—there is nothing to be faithful to, and the control detects it.

Raw generations show three distinct failure modes invisible to a parsed-answer metric. Phi-2 answers plain English prompts correctly but, once `Let's think step by step.` is appended for scoring, produces textbook exposition ("*Use Case 1: Shipping Fragile Items*") that never states a choice. In Vietnamese it emits corrupted scaffold tokens ("*Lê nhiêu*", "*Bnh hi*"). And targets asked in Vietnamese frequently answer correctly *in English* ("*The kayak boat reaches the shore*" for "*Thuyn kayak n b*"), which exact string matching against a Vietnamese gold scores as wrong.

8 Limitations and Ethical Considerations

Single repetition. The original averages three runs; ours are single runs and we report no variance. This is the same gap we identify in the original and we flag rather than repeat it silently.

Missing metric and baselines. We do not report truthfulness, which needs gold explanations that COPA and XCOPA lack; ECQA (Aggarwal et al., 2021) would be required. The original's baselines (Madsen et al., 2024; Wang et al., 2022) are absent from the released code and we did not reimplement them, so comparisons are against published values.

Scope of the controls. Control D measures one redraw, on two of four arms, and only on instances that already flipped; 64.7% is therefore an *upper* bound on stability. The hypothesis that a weak target escapes the ceiling is untested rather than refuted—our intended middle target turned out to be strong (90.5%).

Granularity. Under the released mode S_E carries one bit per instance. A 30-instance estimate of Vietnamese accuracy gave 33.3% with an interval containing chance, against 11.8% at $n = 500$; we twice drew conclusions from $n = 2$ that full runs overturned.

Ethics. All corpora are public. A first run of one explainer had 23% of its calls silently served by a fallback model under rate limiting; we detected this through call accounting, quarantined the run, and fixed the client to wait rather than substitute. We report empty completions from content filtering rather than scoring them as wrong answers. No human subjects were involved.

9 Conclusion

FAITHLM reproduces, and its contrary-hint score is not empty: a real semantic effect survives every control we ran. But the score as reported conflates that effect with how readily the target follows any hint, rates a position-choosing target at 0.755, and loses a third of its flips to a single resampling. Under language transfer the binding constraint is not the metric but the regular-expression reader in front of it. We recommend that intervention-based faithfulness measures report a null-hint baseline, sample the contrary hint $k > 1$ times and take a majority rather than stopping at the first flip, and read the intervened quantity from the model's distribution rather than from its prose.

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